

Running Head: Effects of Load on Peak Power Output Fatigue

**The Effects of Load on Peak Power Output Fatigue
During the Bench Throw**

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ABSTRACT

The ability to create power is an important variable for athletic success. No study to date has compared peak power output (PPO) fatigue across multiple sets and with different loads with the bench throw. This study aimed to begin the process of establishing empirical upper body power training guidelines for moderately strong athletes by determining how load (30%, 45%, 60% 1RM) affects PPO (Watts) dropoff during 3 sets of 10 repetitions of the bench throw. 10 resistance-trained male volunteers ((mean $\pm$ SD): age 20.58 ± 1.36 years, height 176.05 ± 9.09 cm., weight 78.65 ± 9.93 kg, bench press 1RM 99.79 ± 18.52 kg) performed 3 sets of 10 repetitions of the bench throw with one of the three loads during 3 weekly sessions. A Humac 360 device collected concentric phase PPO data during each repetition. The data were analyzed using one-way (Treatment) and two-way (Treatment $\times$ Time) repeated measures ANOVA. A significant decrease in PPO was observed during repetitions 5-7 at 30%, 3-4 at 45%, and 2-3 at 60% 1RM. Based on the results of this study, coaches who want to maximize power should potentially keep sets of upper body plyometrics within these repetition ranges. The authors recommend that moderately strong athletes perform the bench throw on a Smith machine at 45% or 60% 1RM to produce high PPO over multiple sets.

Key Words: Plyometrics, Upper Body, Resistance Exercise, Ballistic

INTRODUCTION

Power, the product of force and velocity, is a key determinant of success in sports (5,16, 22). Plyometrics generate more peak power output (PPO) and mean power than standard resistance training by using the mechanical and neuromuscular properties of the Stretch-Shortening Cycle (SSC), in which elastic energy from the eccentric phase of the movement potentiates the concentric phase (10). Olympic lifts and compound lower body lifts produce significantly more concentric mean power output than upper body movements (30,32). As such, upper body plyometrics are not performed as frequently and have not been investigated as rigorously as lower body plyometrics (7, 28).

Horizontal upper body pressing is a fundamental movement in many sports such as martial arts, football, rugby and wrestling. The bench throw with countermovement (CM) is a plyometric variation of this movement that is more effective than the bench press at stimulating PPO because it maximizes the effect of the SSC. It encourages full acceleration through the concentric phase and thus erases the deceleration observed during the last 40% of the concentric phase during the bench press (11,15,26,32). Incorporating bench throw training with 30-50% one-repetition-maximum (1RM) over a 10-week period significantly increased throwing speed for Major League Baseball players compared to those using the standard bench press (25).

The optimal load for PPO with the upper body has been produced between 30-60% 1RM (3,6,9,12,13,27), although the exact percentage varies considerably within this range. Explanations for these differences include the following: 1) optimal load for PPO may be higher with trained subjects (24); 2) power production will change based on timepoint in periodization

cycle (21). For example, rugby league players who had just completed 6 months of intensive resistance and plyometric workouts including the bench throw in their training produced higher PPO at higher loads versus lower loads (6). PPO is not produced in most cases at the highest loads because peak force is produced at the turnaround of the SSC but peak eccentric velocity occurs earlier in the eccentric phase as the load gets heavier (27), as the athlete creates a more pronounced deceleration effect to stabilize the heavier load (34). Therefore, the inherently faster eccentric stretch speeds possible with lighter loads may increase potentiation through the stretch reflex (12), and stronger athletes may be able to maintain higher eccentric velocities at higher loads to induce this effect.

While the subject of optimal load for power production during one set has been studied extensively, the effect of load on PPO fatigue across multiple sets is an unexplored topic that could yield important training prescription guidelines for strength coaches. To maximize the efficiency and efficacy of their power development training program, coaches need to know how many repetitions cause fatigue across a range of intensities. To the authors' knowledge, there have been no studies comparing PPO fatigue with different loads across multiple sets with the bench throw. Based on anecdotal training evidence, a significant drop is expected between repetitions 3-5 with loads of 30% and 45% 1RM, and at repetition 2 with 60% 1RM (7). The purpose of this study was to begin the process of establishing empirical upper body power training guidelines for moderately strong athletes by determining how load (30%, 45%, 60% 1RM) affects PPO dropoff during 3 sets of 10 repetitions of the bench throw.

METHODS

Experimental Approach to the Problem

Ten athletes attended the Human Performance Laboratory on 4 consecutive Mondays at the same time of day for each subject in an effort to determine differences in PPO during 3 sets (S1,S2,S3) of 10 repetitions of the bench throw, using a within-subjects study design. Loads of 30%, 45%, and 60% 1RM were chosen as these elicit the highest PPO during the bench throw (6, 12). The first session was 1RM bench press testing, and sessions 2-4 were power testing with the bench throw using a Humac 360 linear position transducer (LPT) (Computer Sports Medicine Inc., Stoughton, MA). Although there is not as much research into reliability of power output from LPTs with the bench throw compared to CM jumps (18), in a prior study PPO was statistically similar (ICC from .92-.96) in 4 sessions spaced at least 48 hours apart during 30 repetitions of the bench throw at 30% 1RM, demonstrating reliability and the lack of a learning effect (2). ICC's for peak velocity for a set of 15 repetitions of bench throw at 30, 40, and 50% 1RM had ICC's of .97, .98, and .97 respectively, showing very high reliability (14). Recording the velocity of hand movement during punching trials, the Humac 360 had an ICC of 0.9982 (95% CI = 0.960-.993) (20).

A pilot study (N=5) was conducted in which PPO was measured during 1 set of 10 repetitions of the bench throw at 30% 1RM, to ensure that equipment and software were reliable and to practice the familiarization protocol. As there was a significant drop in PPO at repetition 9 during 30% 1RM bench throws during the pilot study, 10 repetitions was chosen for the current study to increase the chance that all subjects would reach significant fatigue with the lightest load, 30% 1RM, while not creating too much fatigue that might affect subsequent sets.

Subjects

Ten male NCAA Division I University lacrosse athletes volunteered for this study (Table 1). None of the subjects had performed the bench throw exercise before the study. They had no current upper body injuries and had not undergone any rehabilitation for such an injury within the 6 months prior to the first session. All subjects were informed of the purpose, procedures, benefits and risks of the investigation prior to signing the IRB-approved informed consent form and medical history questionnaire. The SSU Institutional Review Board approved the use of human subjects for this study.

Table 1. Subject variables (N=10).

[Place Table 1 here]

Procedures

Subjects abstained from any upper body resistance or plyometric training for at least 72 hours prior to each session. In a previous study, PPO was equivalent between sets when athletes were given a rest of 2 or 3 minutes during bench throws at 40% 1RM (19). However, to ensure that rest would be adequate at the heavier load of 60% 1RM, the rest interval between sets for the current study was fixed at 3 minutes. Previous research has shown that 3 minutes is needed for ATP replenishment and flushing out fatigue-inducing glycolytic products (1,35).

Session 1 consisted of anthropometric measurements followed by a 1RM bench press test and a familiarization session with the bench throw movement. Bench press 1RM was assessed on

a Smith machine (Cybex International, Model 5341-001-97, Owatonna, MN) using the National Strength and Conditioning Association 1RM testing guidelines (4). After a 3 minute rest and demonstration, the bench throw familiarization protocol was 1 set of 5 repetitions each at 30%, 45%, and 60% 1RM, in order of ascending load, with 3 minutes rest between sets.

Sessions 2-4 consisted of 3 sets of 10 repetitions of bench throw PPO testing using the Smith machine, after a warmup consisting of 10 repetitions with the unloaded bar (11 pounds) (Figure 1). The loads were randomized and balanced across the three testing sessions. The Humac 360 was attached to the end of the barbell and the encoder was placed on the floor directly underneath the retractable belt. The unit collected PPO during the concentric phase of each repetition. For the duration of every set participants assumed a 5-point body contact position (2 feet, hips, shoulders, head) and gripped the bar with a closed, pronated grip with which they felt the most comfortable and this grip was replicated on all trials. Subjects started every repetition from a locked out position to insure that every repetition included an eccentric phase. They were instructed to lower the barbell without touching the chest for safety reasons. During each repetition they were verbally encouraged to forcefully expel the barbell from their hands in order to minimize turnaround from the eccentric to concentric phase. In a previous study, minimizing the length of the amortization phase resulted in the greatest power output when compared to slow eccentric phases and isometric pauses in the bottom position (29).

Figure 1. Sessions 2-4: Peak power output testing for the bench throw. Sessions were separated by 7 days.

[Place Figure 1 here]

Statistical Analyses

All data were analyzed using the statistical package SPSS 22 (SPSS Inc., Chicago, IL) and is presented as mean $\pm$ SD. Subject characteristics are presented as descriptive statistics for age, height, body mass, and strength measurements. Measures of central tendency and variations were calculated for all variables, and outliers (> 2 SD) were identified and analyzed for confounding factors. There were 5 data values that were deemed to be statistical outliers as the result of confounding variables and thus were excluded from the analysis. Peak power was calculated as the loaded weight (Force) $\times$ peak instantaneous velocity of the concentric phase. The data were analyzed using one-way (Load) and two-way (Load $\times$ Repetition) analysis of variance, with repeated measures ANOVA. In the event of a significant F score, Fisher's LSD and/or paired t-test was used *post hoc* to determine pairwise differences. Significance was set at $p < 0.05$. Percent decline was calculated for each repetition within a set as $\{(\text{Highest PPO in Set} - \text{Repetition's PPO}) / \text{Highest PPO} \times 100\}$.

RESULTS

Peak power of each repetition as a percentage of maximal PPO within each set is presented in Figure 2. Repeated measures ANOVA comparing the highest PPO within each set to the 9 other repetitions indicated the point at which significant ($p \leq .05$) PPO fatigue occurred. During 30% 1RM testing, power output significantly declined at repetition 7 for S1, 5 for S2, and 5 for S3; during 45% 1RM at repetition 4 for S1, 4 for S2, and 3 for S3; and during 60% 1RM at repetition 2 for S1, 3 for S2 and 3 for S3.

Figure 2. Peak Power Output by Repetition (Percentage of Max)

[Place Figure 2 here]

Figure 3 depicts the average PPO by repetition, compared across loads. By repetition 2, 60% 1RM is lower than 45% 1RM, and a similar shift occurs by repetition 5, where 60% 1RM begins to produce less PPO than 30% 1RM. At 45% 1RM, the athletes were able to produce and maintain higher PPO compared to lower and higher intensities.

Figure 3. Average PPO by Repetition

[Place Figure 3 here]

DISCUSSION

The primary finding of the present study was that significant PPO fatigue during the bench throw occurred at repetitions 5-7 using 30% 1RM, at repetitions 3-4 using 45% 1RM, and at repetitions 2-3 using 60% 1RM. These repetitions mark the point at which PPO dropped significantly compared to the repetition with the highest power within that set. Maximal PPO within each set was not defined by the first repetition in many cases because the first repetition did not include the additional velocity of the barbell falling that every subsequent rep used. This method has been used by Garcia-Ramos et al. (14) to more accurately compare PPO within sets.

Based on training experience and anecdotal evidence, Baker and Newton (7) suggested specific repetition ranges for power training: 8-10 repetitions at 25% 1RM, 3-5 repetitions at 25-45% 1RM, and 2-3 repetitions at 45-55% 1RM. The results of the current study corroborate these

suggestions and provide empirical evidence for them. In a previous investigation with advanced athletes using $45\% \pm 6\%$ 1RM, there was a significant decline in PPO by the 6th repetition of the bench throw, and a 11.2% in PPO by repetition 10 (8). However, athletes in the current study fatigued at repetition 3-4 and suffered a 14.78% decline in PPO by repetition 10. These differences are possibly due to these athletes having no previous training with the bench throw, and relatively low absolute strength (99.79 ± 18.52 kg) compared to advanced strength-power athletes. Another investigation, using 40% 1RM, demonstrated a significant decline in PPO by repetition 3 and 4 during the bench throw with athletes similar in absolute strength level to the current study (19), corroborating the fatigue at 3-4 reps at 45% 1RM in the current study.

Knowing the optimal load for power training for a specific athlete and movement is important because the highest PPO attained during the concentric phase of a plyometric movement is a key variable for athletic success (5). Suchomel, Comfort and Stone (31) make the crucial distinction that the optimal load is only for a given athlete and movement, and is highly subject to fatigue state. Forty-five percent 1RM produced significantly greater average PPO across the 3 sets of 10 repetitions compared to both 60% and 30% 1RM. The optimal load was previously shown to be 30% 1RM for athletes who bench $1.1 \times$ BW (bodyweight) (33), and 46-62% for athletes who can bench press $1.4 \times$ BW (6). As subjects in the current study's average relative bench press strength was $1.3 \times$ BW, attaining maximal PPO at 45% 1RM corroborates these findings.

When the sets were calculated as only containing the repetitions up to the point of fatigue, 45% and 60% 1RM were no different except for 60% 1RM set 3, which was

significantly lower than all the other sets at 45% and 60% 1RM. 30% 1RM was significantly lower than all sets at 45 and 60%. This analysis shows the clear superiority of 45% 1RM if the goal is producing and maintaining high power output with the bench throw over multiple sets. A limitation is the fact that the athletes performed repetitions past fatigue in each set which might have caused unwanted residual fatigue in subsequent sets. Given this situation, 3 minutes may have been insufficient for the sets at 60% 1RM. However, in a previous study, PPO was equivalent between sets when athletes were given a rest of 2 or 3 minutes during bench throws at 40% 1RM (19). In addition, the coefficient of variation (CV) values in the current study were very low (.85-2.8%) when comparing the first 2 reps of every set at each intensity, demonstrating low variability.

Coaches should consider that low-load and high-load velocity-based training are required in tandem to increase both aspects of an athlete's force-velocity curve. In addition, training loads should correspond to whether force or velocity is more important for the given sport, training session and periodization timepoint. Because fatigue occurs after so few repetitions at 60% 1RM, coaches may want to consider the use of cluster sets with upper body power training. When comparing cluster sets to traditional continuous repetitions, significant PPO dropoff occurred at later repetitions with 6RM bench press (23), and peak velocity was likewise higher with bench throws (14) and clean pulls (17).

Future research could investigate the effects of performing multiple sets with the repetition ranges identified here as causing significant fatigue. This would reveal the optimal number of sets for an empirical volume prescription for power training. Comparing the power

outputs from different athletes using the same repetition ranges discovered in the current investigation would give insight into the effect of strength level and previous experience with the modality on PPO and fatigue.

PRACTICAL APPLICATIONS

Based on the results of this study, the authors recommend that moderately strong athletes learning upper body plyometrics perform the bench throw on a Smith machine at 45% or 60% 1RM to produce high PPO over multiple sets. Coaches who want to maximize power when training athletes from this population should potentially keep sets of upper body plyometrics at 30% 1RM to 5-7 repetitions, 45% 1RM to 3-4 repetitions, and 60% 1RM to 2-3 repetitions.

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TABLES

Table 1. Subject variables (N=10).

Variables	Age (y)	Height (cm)	Body Weight (kg)	Bench Press Training Duration (y)	1-RM Bench Press (kg)
Mean	20.58	176.05	78.65	4.95	99.79
SD	1.36	9.09	9.93	2.23	18.52

FIGURES

Figure 1. Sessions 2-4: Peak power output testing for the bench throw. Sessions were separated by 7 days.

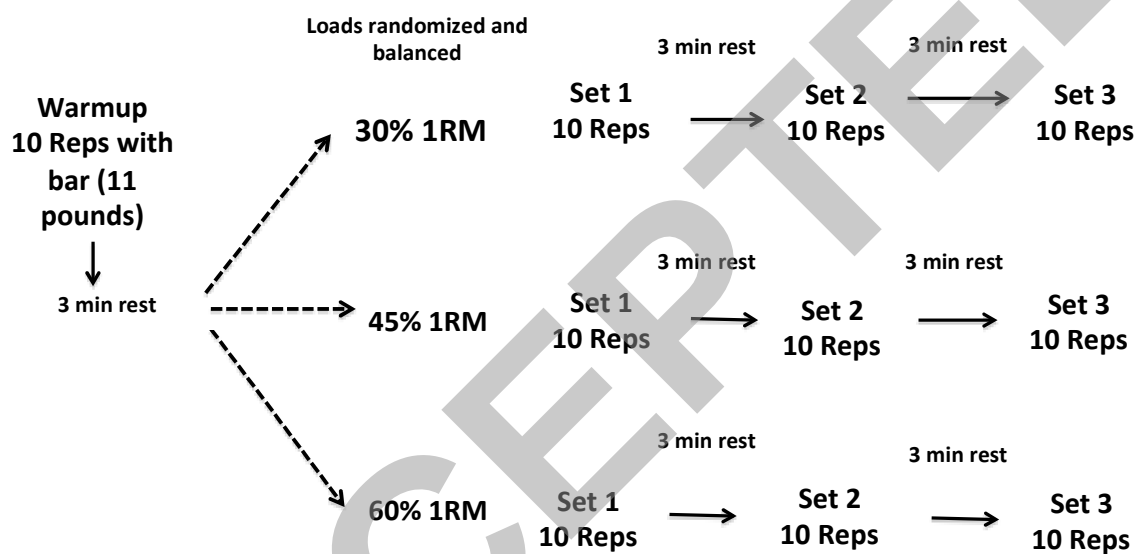


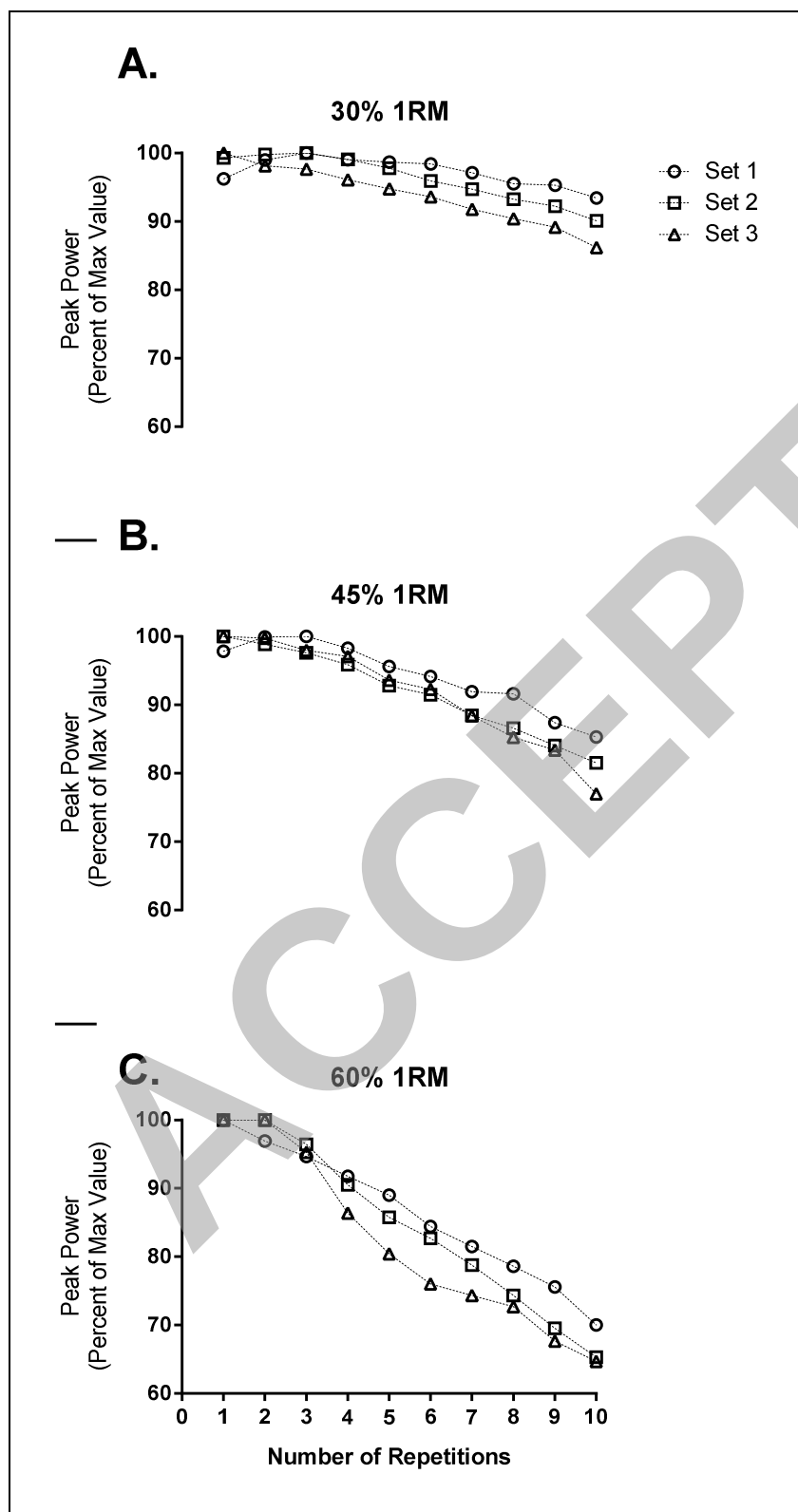
Figure 2. Peak Power Output by Repetition (Percentage of Max)

Figure 3. Average PPO by Repetition